

AI Usage Note

Project Title

AI-Powered Ticket Management MCP Server

AI Tools Used

- Claude AI
- Gemini (Antigravity)
- ChatGPT

Purpose2 of AI Usage

AI tools were used to accelerate development, generate boilerplate code, design system architecture, create documentation, and assist in testing. All AI-generated content was manually reviewed, modified, and validated by the team before inclusion in the final project.

Areas Where AI Assisted

Architecture Design

AI helped design the layered architecture consisting of:

- MCP Server Layer
- Service Layer
- Database Layer
- Agent Client Layer

Database Design

AI assisted in creating the SQLite database schema for:

- Tickets
- Comments
- Knowledge Base Articles

MCP Server Development

AI helped generate and refine MCP tools:

- list_tickets
- get_ticket
- create_ticket
- add_comment
- search_kb
- update_ticket_status
- resolve_ticket

Testing

AI assisted in generating pytest test cases, validation scenarios, and debugging support.

Documentation

AI helped create:

- README.md
- Architecture documentation
- API documentation
- Sample data files
- Setup instructions

What AI Got Wrong

During development, AI occasionally generated incorrect import paths, incomplete MCP configurations, and tool schemas that required manual correction. These issues were identified and fixed by the team through testing and validation.

Best Prompts Used

1. Design an MCP-based Ticket Management System using Python and SQLite.
2. Generate SQLite schema for tickets, comments, and knowledge base articles.
3. Implement MCP tools for ticket lifecycle management.
4. Create pytest test cases for MCP tools and services.
5. Generate sample input and output files for project demonstration.

Human Contributions

- Requirement analysis
- Feature selection and design decisions
- Claude Desktop integration
- MCP server debugging
- Testing and validation
- GitHub repository management
- Final documentation and presentation

Conclusion

AI was used as a development assistant to improve productivity and reduce development time. However, all implementation decisions, testing, integration, validation, and final project ownership remained with the team.